



Project : Wazuh Deployment & Windows AD Monitoring

Author: Khandaker Mahidul Islam

Wazuh Deployment & Windows AD Monitoring (Step-by-Step)

1. Lab Pre-Check

- **Wazuh Server (OVA):** 192.168.65.147
- **Windows Server 2022 (AD/DC):** 192.168.65.146
- On Wazuh Server, opened syslog port:
`sudo ufw allow 514/udp`
- Synced both systems with NTP to avoid event timestamp mismatch.

Wazuh Server Start : (Command)

- `sudo systemctl start wazuh-manager`
- `sudo systemctl start wazuh-dashboard`
- `sudo systemctl daemon-reload`
- `sudo systemctl start wazuh-indexer`

IF any Problem Wazuh manager Start again, then try this command:

Wazuh Manager restart

- `sudo systemctl stop wazuh-manager`
- `sudo pkill -f wazuh-remoted`
- `sudo systemctl start wazuh-manager`
- `sudo systemctl status wazuh-manager`

2. Wazuh Manager Configuration

- Opened the Wazuh Manager config file:
- `sudo nano /var/ossec/etc/ossec.conf`
- Added syslog receiver block:
- `<remote>`
- `<connection>syslog</connection>`
- `<port>514</port>`
- `<protocol>tcp</protocol>`
- `<allowed-ips>192.168.65.146</allowed-ips>`
- `</remote>`

- (Optional) Added secure agent connection block:
 - `<remote>`
 - `<connection>secure</connection>`
 - `<port>1514</port>`
 - `<protocol>tcp</protocol>`
 - `</remote>`
 - Restarted manager:
 - `sudo systemctl restart wazuh-manager`
 - `sudo systemctl status wazuh-manager -l`
-

3. NXLog on Windows AD/DC

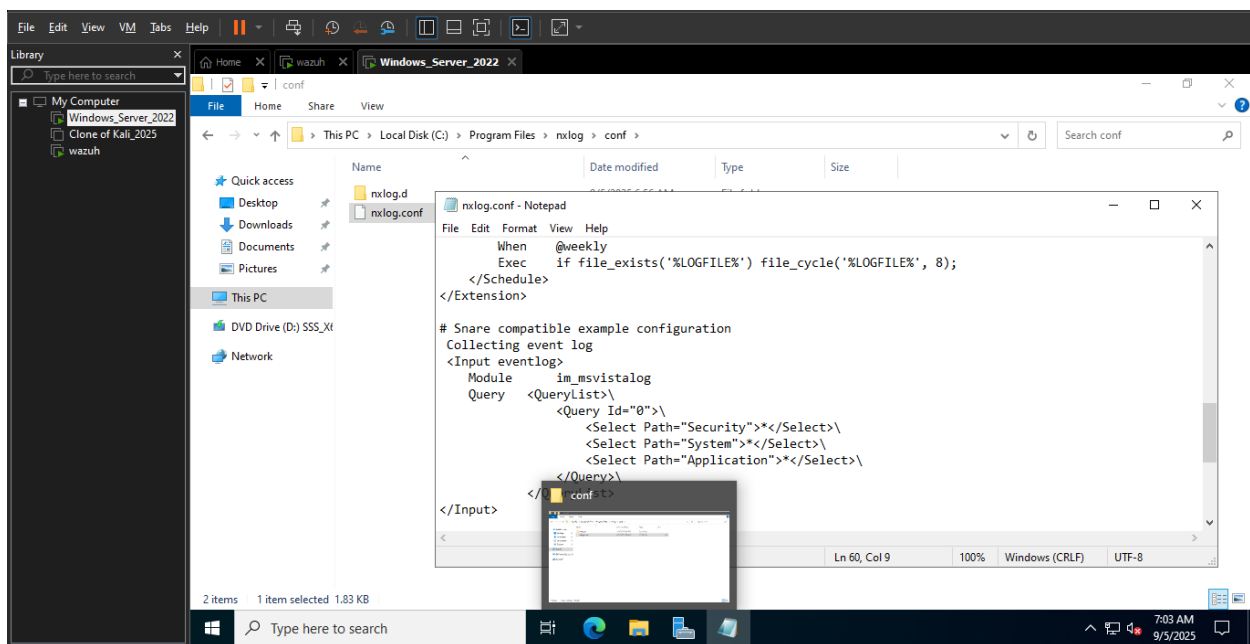
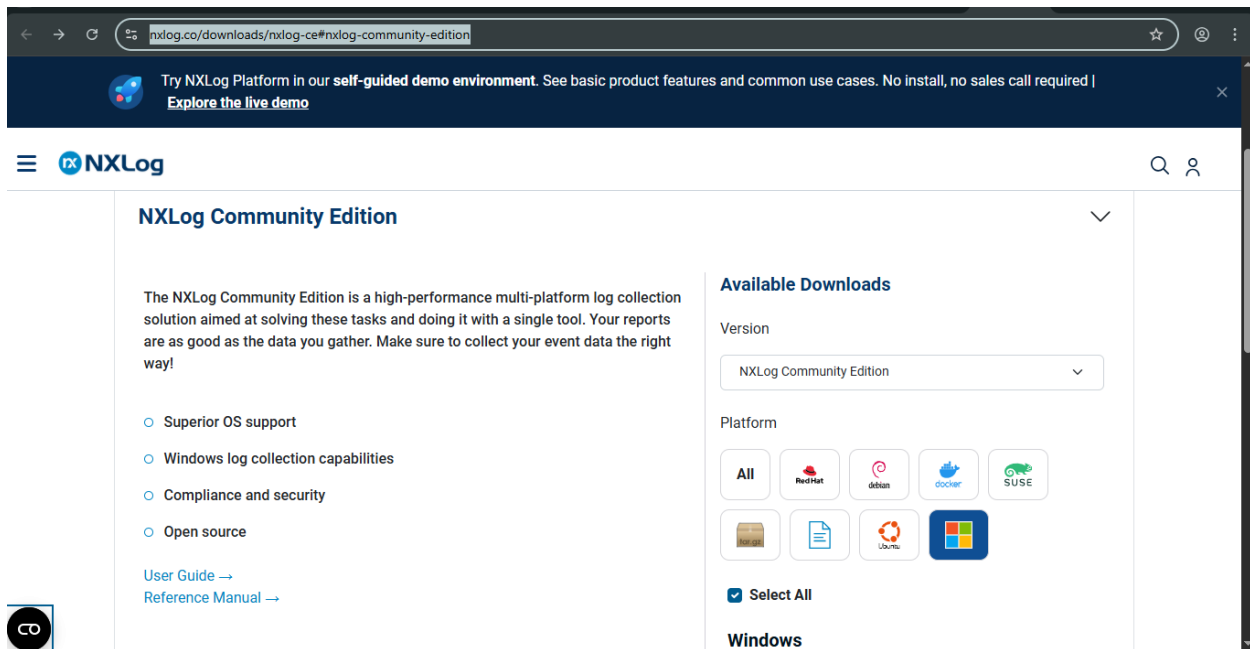
- Installed **NXLog Community Edition**.
- Config file path: `C:\Program Files\nxlog\conf\nxlog.conf`

3.1 Forward all Security Logs (basic)

```
<Input security_evt>
  Module im_msvistalog
  Query <QueryList>\
    <Query Id="0">\
      <Select Path="Security">*</Select>\
    </Query>\
  </QueryList>
</Input>

<Output out_json>
  Module om_udp
  Host 192.168.65.147 # Wazuh Manager IP
  Port 514
  Exec to_json();
</Output>

<Route 1>
  Path security_evt => out_json
</Route>
```



3.2 Forward only important EventIDs (optimized)

Example filter for login/logout, lockouts, account changes:

```
<Select Path="Security">*[System[(EventID=4624 or EventID=4625 or
EventID=4634 or EventID=4648 or EventID=4672 or EventID=4740 or EventID=4720
or EventID=4722 or EventID=4725 or EventID=4768 or EventID=4769 or
EventID=4776)]]</Select>
```

- Restart NXLog service:

- net stop nxlog
 - net start nxlog
-

4. Local Rules on Wazuh Manager

sudo nano /var/ossec/etc/rules/local_rules.xml

Created custom rules for AD security events:

```
<group name="windows,auth,syslog,json,">
  <rule id="100001" level="10">
    <field name="EventID">4625</field>
    <description>Windows failed logon</description>
    <group>authentication_failed,windows,</group>
  </rule>

  <rule id="100002" level="12">

    <field name="EventID">4740</field>
    <description>AD account locked</description>
    <group>account_lockout,windows,</group>
  </rule>

  <rule id="100003" level="12">
    <frequency>5</frequency>
    <timeframe>120</timeframe>
    <description>Multiple failed logons</description>
    <group>bruteforce,windows,</group>
  </rule>
</group>
```

- Restarted manager:
 - sudo systemctl restart wazuh-manager
-

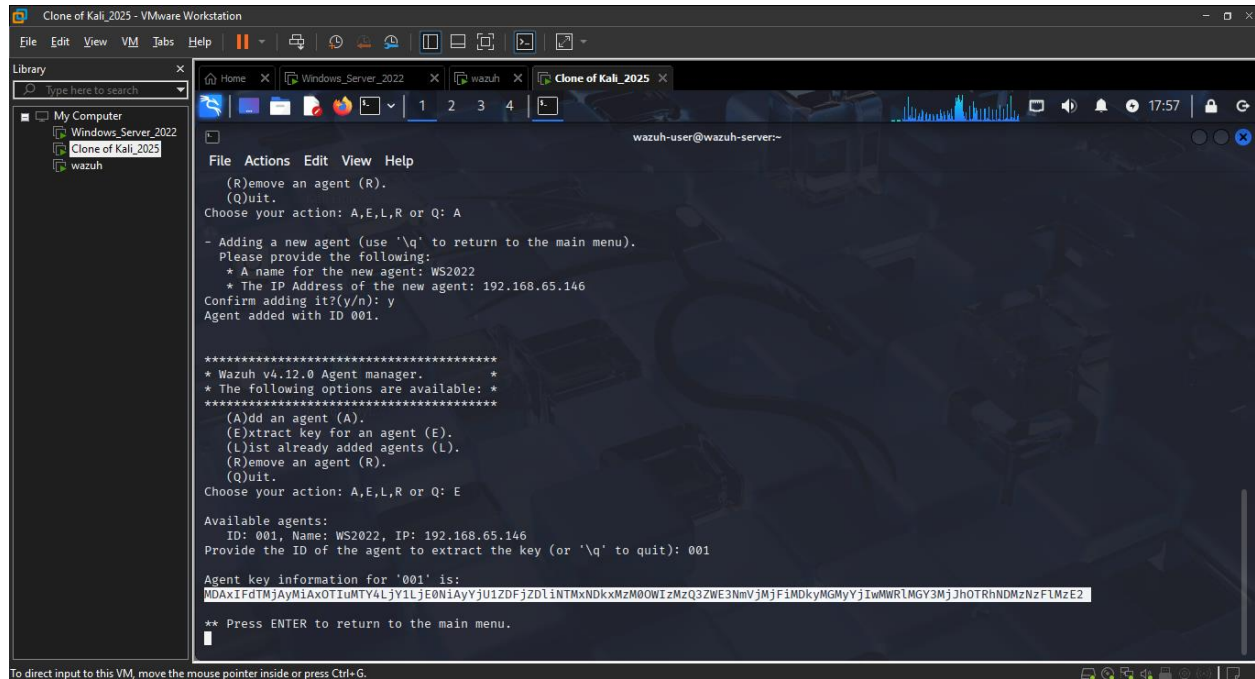
5. Wazuh Agent Installation (Windows AD/DC)

5.1 Add Agent on Wazuh Manager

- Run:
- sudo /var/ossec/bin/manage_agents
- **Select A) Add agent**
- Enter:
 - Name: ws2022
 - IP: 192.168.65.146 [windwos-server-ip]
- Confirm → Agent ID 001 created.

5.2 Extract Key

- From the same menu, choose **Extract key** → select Agent ID 001.
- Copy the generated **auth key**.



5.3 Configure Agent on Windows

- Install Wazuh Agent on Windows.
- Edit:
 - C:\Program Files (x86)\ossec-agent\ossec.conf
- Update server IP and auth key:
 - <server>
 - <address>192.168.65.147</address> [wazuh manager ip]
 - <port>1514</port>
 - </server>
 -
 - <auth>
 - <key>PASTE_AUTH_KEY_HERE</key>
 - </auth>
- Save and restart service:
 - net stop WazuhAgent
 - net start WazuhAgent

5.4 Verify Connection

- On Wazuh Manager:
 - sudo /var/ossec/bin/agent_control -l

- Output should show `WS2022` as **Active**.

Wazuh Agent Installation (Windows AD/DC)

5.1 Add Agent on Manager

- Used `manage_agents` to add a new agent (e.g., WS2022 at 192.168.65.146).
- Extracted the **auth key** for this agent.

5.2 Configure Windows Agent

- Installed the Wazuh Agent MSI on Windows AD/DC.
 - Updated `ossec.conf` with:
 - Wazuh Manager IP (192.168.65.147).
 - The extracted **auth key**.
 - Restarted the agent service.
-

6. Verify Agent Installation (Windows)

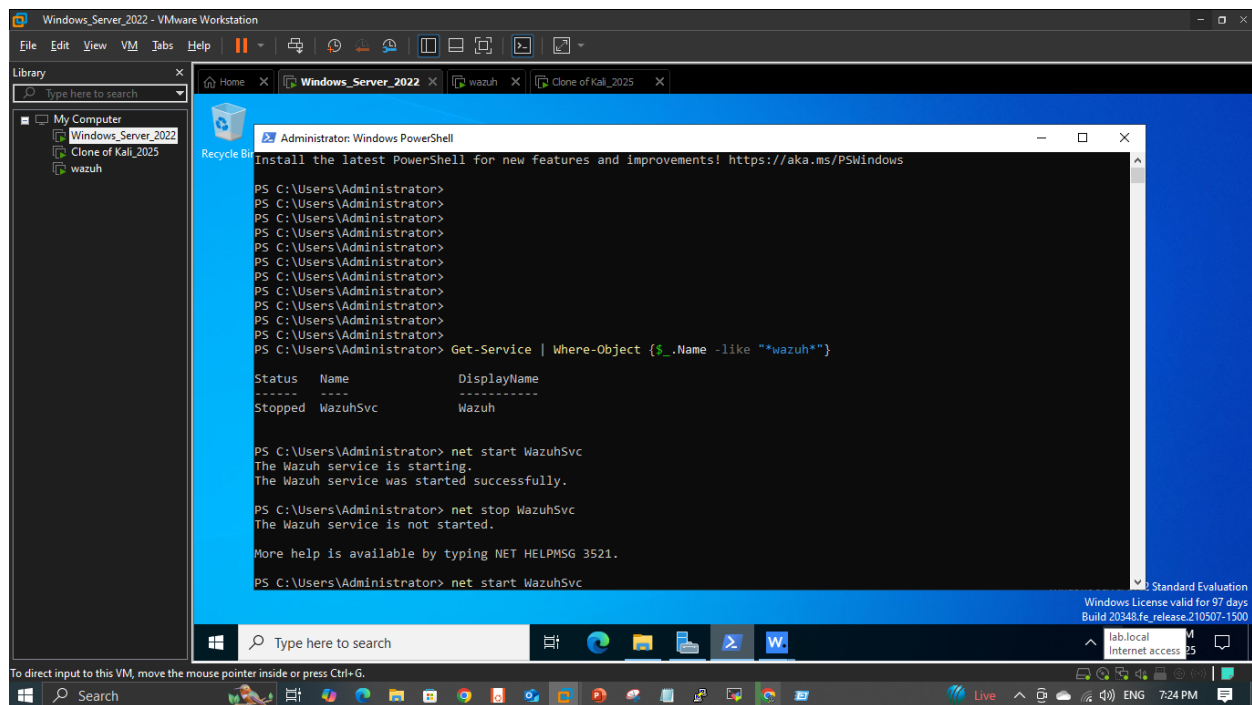
6.1 Check Installation Directory

- In Windows Explorer:
- `C:\Program Files (x86)\ossec-agent`
- If this folder does not exist → Agent is not installed.

6.2 Check Service Name

- Open PowerShell (Admin):
- `Get-Service | Where-Object {$_.Name -like "*wazuh*"}`
- Example output:

Name	Status
WazuhAgent	Running
-  Note: The correct service name is usually **WazuhAgent**, not `wazuh-agent`.



6.3 Start / Restart Service

- Start Agent:
- `net start WazuhAgent`
- Check status:
- `sc query WazuhAgent`
- Restart Agent:
- `net stop WazuhAgent`
- `net start WazuhAgent`

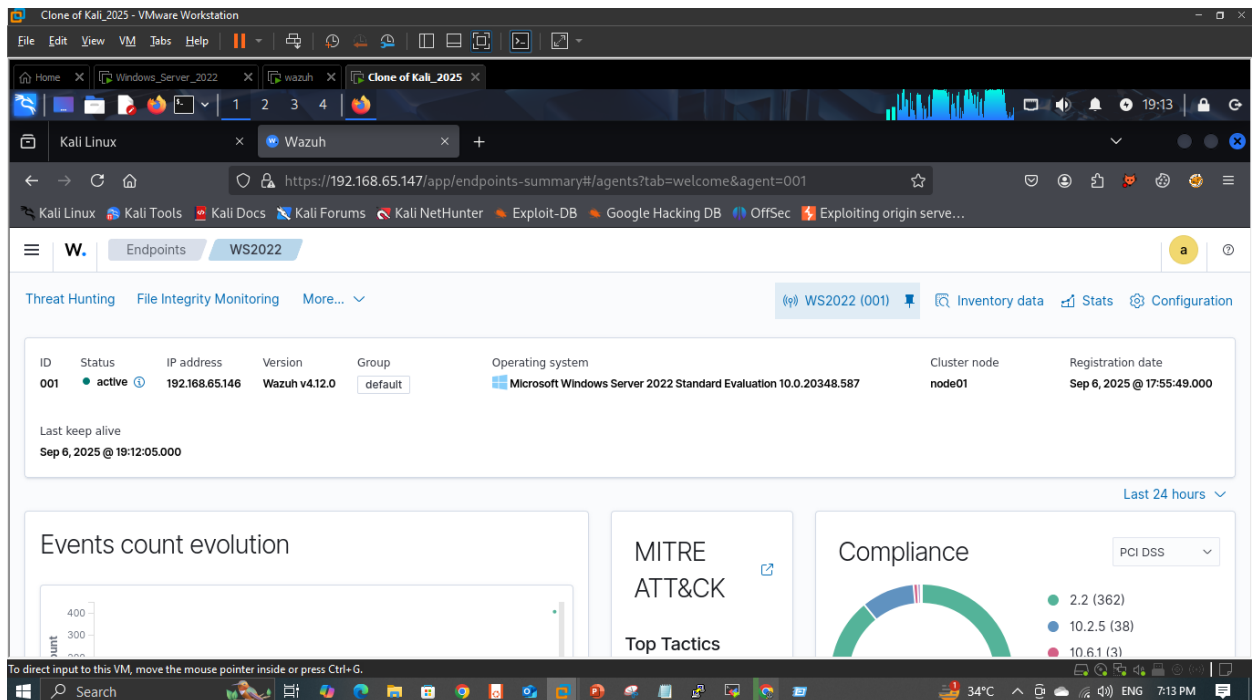
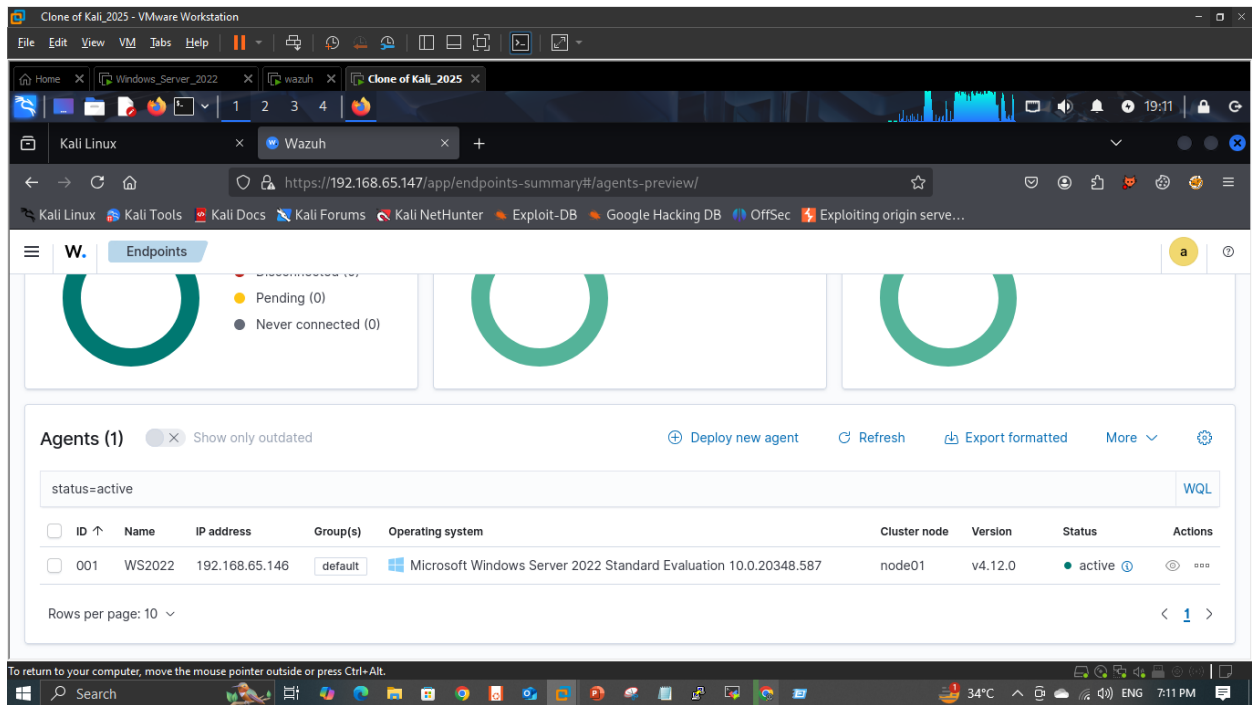
6.4 If Agent Not Installed

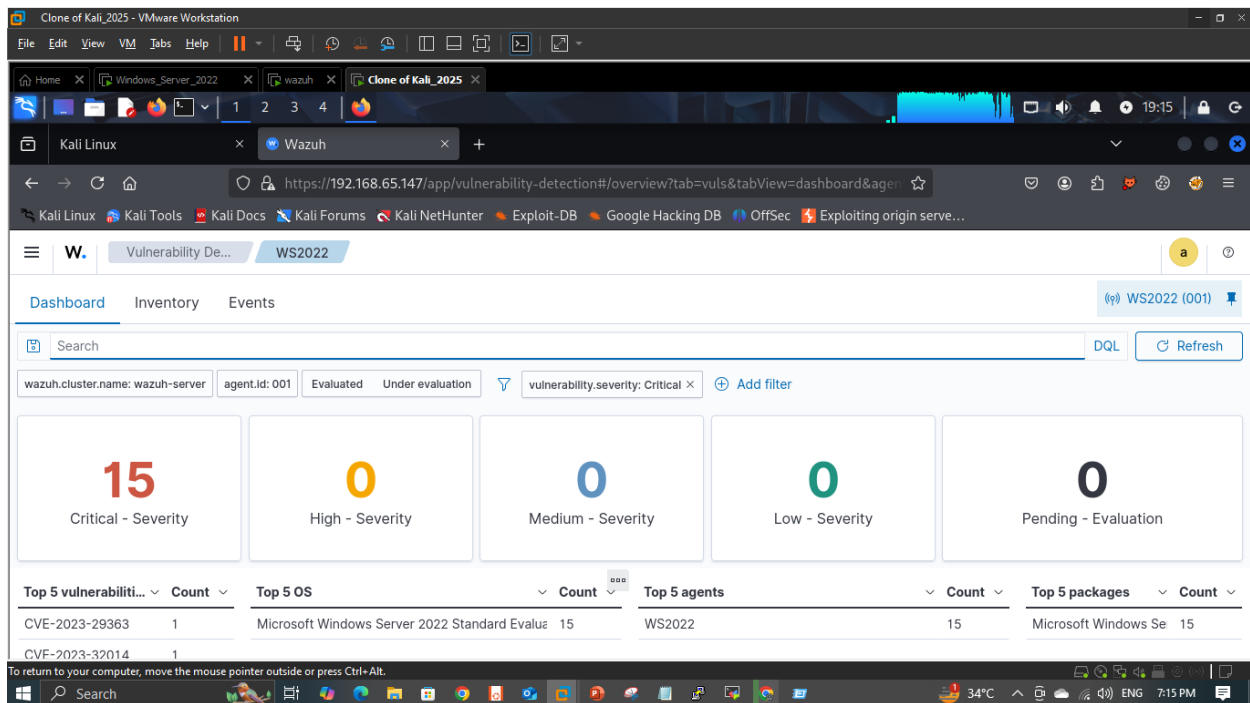
- Download the **Wazuh Windows Agent MSI** from official Wazuh site.
- Run as Administrator → Enter Wazuh Manager IP → Finish.
- After installation, the **WazuhAgent** service will be created automatically.

6. Monitoring in Wazuh Dashboard

6.1 Access Dashboard

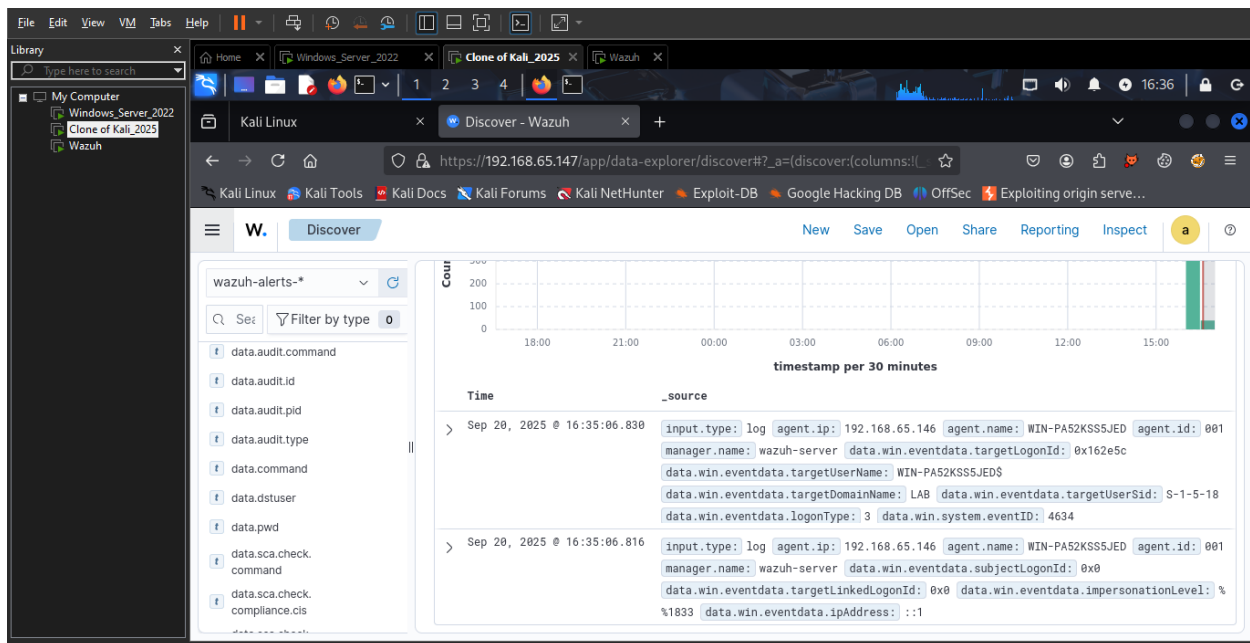
- URL:
- <https://192.168.65.147>





6.2 Security Events

- Navigate to **Security Events / Discover**.
- Index used: `wazuh-alerts-*`.
- Use filters or queries to search events.



Clone of Kali_2025 - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- Windows_Server_2022
- Clone of Kali_2025
- Wazuh

Kali Linux x Wazuh x +

https://192.168.65.147/app/vulnerability-detection#/overview?tab=vuls&tabVie

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Exploiting origin serve...

W. Vulnerability De... WIN-PA52KSS5JED

Dashboard Inventory Events WIN-PA52KSS5JED (001)

Search DQL Refresh

wazuh.cluster.name: wazuh-server agent.id: 001 Evaluated Under evaluation vulnerability.severity: Critical Add filter

15 hits

Export Formatted 48 available fields Columns Density Sort fields Full screen

agent.name	package.name	package.version	vulnerability.descrip...	vulnerability.severity	vulnerability.id
WIN-PA52KSS5JED	Microsoft Windows Serv...	10.0.20348.587	Windows OLE Remote C...	Critical	CVE-2025-21298
WIN-PA52KSS5JED	Microsoft Windows Serv...	10.0.20348.587	Windows Lightweight Di...	Critical	CVE-2024-49112
WIN-PA52KSS5JED	Microsoft Windows Serv...	10.0.20348.587	Windows KDC Proxy Re...	Critical	CVE-2024-43639
WIN-PA52KSS5JED	Microsoft Windows Serv...	10.0.20348.587	Windows Reliable Multic...	Critical	CVE-2024-38140
WIN-PA52KSS5JED	Microsoft Windows Serv...	10.0.20348.587	Windows Netlogon Elev...	Critical	CVE-2024-38124

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Kali Linux x Discover - Wazuh x +

https://192.168.65.147/app/data-explorer/discover#?_a=(discover:(columns:!!_source),isDirty:!!_sort)

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Exploiting origin serve...

W. Discover New Save Open Share Reporting Inspect

wazuh-monitoring-* Search Filter by type 0

Selected fields

- _source

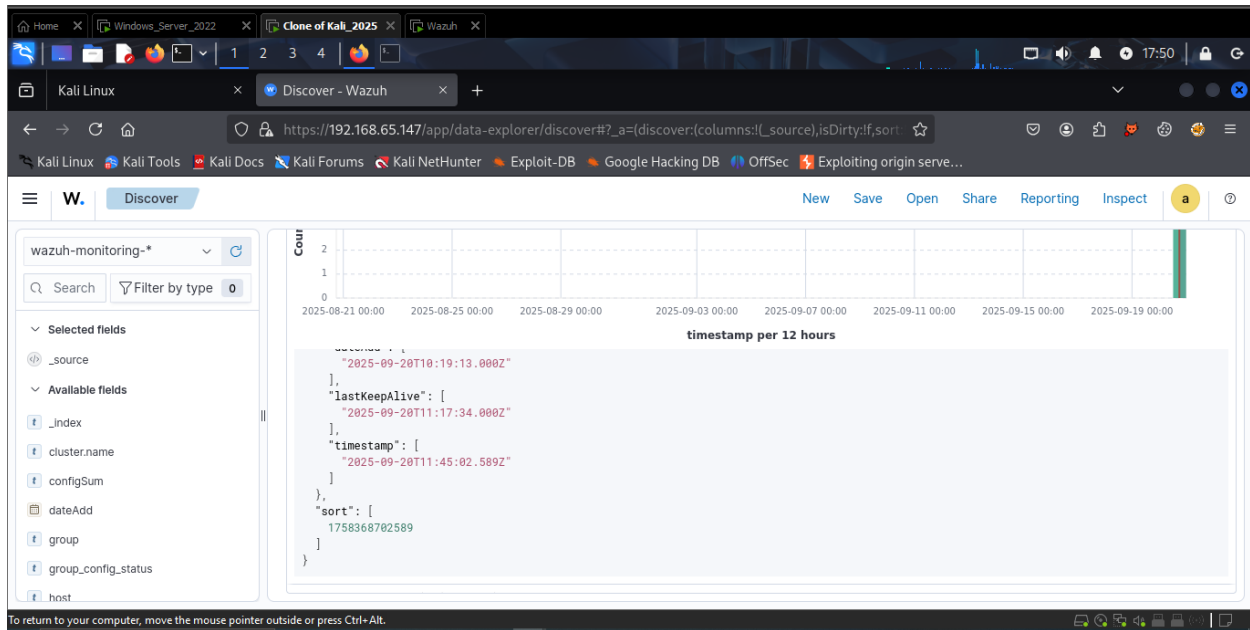
Available fields

- _index
- cluster.name
- configSum
- dateAdd
- group
- group_config_status
- host

Count

timestamp per 12 hours

```
{
  "_index": "wazuh-monitoring-2025.38w",
  "_id": "-8nxZpkBtBTqUqYIN-B",
  "_version": 1,
  "_score": null,
  "_source": {
    "os": {
      "build": "20348.587",
      "major": "10",
      "minor": "0",
      "name": "Microsoft Windows Server 2022 Standard Evaluation",
      "platform": "windows",
      "uname": "Microsoft Windows Server 2022 Standard Evaluation",
      "version": "10.0.20348.587"
    }
  }
}
```



The screenshot shows the Wazuh Threat Hunting interface. The table displays events for agent WIN-PA52KSS5JED. The table has 775 hits and shows events from Sep 19, 2025 @ 17:53:30.422 to Sep 20, 2025 @ 17:53:30.422. The table is sorted by 1 field.

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 20, 2025 @ 17:52:36.3...	WIN-PA52KSS5JED	Wazuh agent disconnected.	3	504
Sep 20, 2025 @ 17:17:07.3...	WIN-PA52KSS5JED	Windows User Logoff	3	60137
Sep 20, 2025 @ 17:17:07.3...	WIN-PA52KSS5JED	Windows Logon Success	3	60106
Sep 20, 2025 @ 17:16:33.5...	WIN-PA52KSS5JED	Windows User Logoff	3	60137
Sep 20, 2025 @ 17:16:22.8...	WIN-PA52KSS5JED	Windows Logon Success	3	60106
Sep 20, 2025 @ 17:16:07.3...	WIN-PA52KSS5JED	Windows User Logoff	3	60137
Sep 20, 2025 @ 17:16:07.3...	WIN-PA52KSS5JED	Windows Logon Success	3	60106

6.3 Useful Queries (DQL)

- **Failed logins:**
- winlog.event_id:4625
- **Successful logins:**
- winlog.event_id:4624
- **Account lockouts:**
- winlog.event_id:4740

- **Specific user activity:**
- `user.name:"administrator"`

6.4 Sort & Export

- Sort by **@timestamp** to see **first login / last logout**.
 - Save queries for daily monitoring.
 - Export results as CSV/PDF for HR/security reporting.
-

Outcome

- Wazuh Manager is receiving **Windows AD Security logs** via NXLog.
- Wazuh Agent is **registered and active**.
- Local rules trigger alerts for:
 - Failed logins,
 - Account lockouts,
 - Brute force attempts.